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1 Power Query Editor

Power Query Editor is the data preparation and transformation engine in Microsoft Power BI. It enables users to connect to multiple data sources, clean and reshape data, combine datasets, remove inconsistencies and prepare high-quality data before loading it into the Power BI data model.

Almost every professional Power BI project begins with Power Query because clean and structured data forms the foundation of reliable reports, dashboards, and DAX calculations.

This chapter introduces the Power Query Editor interface, its components, common transformations, query management, and best practices used in enterprise Business Intelligence (BI) projects.

Definition:

Power Query Editor is Microsoft's Extract, Transform, and Load (ETL) tool available in Power BI Desktop. It enables users to prepare raw data before it is loaded into the data model.

Power Query follows the ETL process:

- Extract data from one or more sources.
- Transform the data.
- Load the cleaned data into Power BI.

Power Query records every transformation as a reusable step, allowing automatic execution during every refresh.

Problems

1.1 Open the Power Query Editor for an imported Sales dataset and identify its major interface components

Solution:

- Open Power BI Desktop
- Import the Sales dataset
- Click Home to Transform Data
- Observe the Query Editor window
- identify:
 - Ribbon
 - Query Pane
 - Data Preview
 - Formula Bar
 - Query Setting
 - Applied Steps

1.2 Locate the Query Pane and identify all imported queries

Solution:

- Open Power Query Editor
- Locate the Query Pane on the left side
- Review every imported query
- Select each query individually
- Observe the data preview for each query

1.3 Display the Formula Bar and examine the transformation formula generated by Power Query.

Solution:

- Open Power Query Editor
- Select the View tab
- Enable the Formula Bar
- Select any Applied step
- Observe the generated M expression

1.4 Review the Applied Steps for a query after performing several transformations.

Solution:

- Open Power Query Editor
- Select a query
- Remove one unnecessary column
- Change a data type
- Rename a column
- Review the Applied Steps pane

1.5 Use the Data Preview area to inspect imported data before applying transformation

Solution:

- Open Power Query Editor
- Select the Customer query
- Review the Data preview
- Verify:
 - Column names
 - Data types
 - Blank values
 - Duplicate rows
 - Missing records

1.6 Rename imported queries to meaningful business names

Solution:

- Open Power Query Editor
- Locate the Query Pane
- Right-click the query
- Select Rename
- Rename the queries as:
 - Sales
 - Customers
 - Products
 - Stores
 - Calendar

1.7 Display the Query Setting pane and identify its sections

Solution:

- Open Power Query Editor
- Click the view tab
- Ensure Query Settings is enabled
- Observe:
 - Properties
 - Applied Steps

1.8 Refresh the Data preview after modifying the source dataset

Solution:

- Open Power Query Editor
- Select the desired query
- Click Home then Refresh Preview
- Wait for the preview to reload
- Verify the latest records

1.9 View the complete transformation script using the Advanced Editor

Solution:

- Open Power Query Editor
- Select a query
- Click Home then Advanced Editor
- Review the complete M script
- Close the editor without making changes

1.10 Close Power Query Editor and apply all completed transformations to the Power BI data model

Solution:

- Verify all Applied Steps
- Save the query
- Click Home then Close and Apply
- Wait until Power BI loads the transformed data
- Verify that the tables appear in the Data and Model views.

1.11 View the Query Dependencies diagram to understand how multiple queries are related.

Solution:

- Open Power Query Editor
- Select the view tab.
- Click Query Dependencies.
- Observe the dependency diagram.
- Identify source queries and dependent queries.

1.12 Create a duplicate of an existing query for independent modifications.

Solution:

- Open Power Query Editor.
- Right-Click the Sales Query.
- select duplicate
- Rename the new query as Sales Summary
- Verify that both queries exist independently.

1.13 Create a reference query from an existing Sales query.

Solution:

- Open Power Query Editor.
- Right-click the Sales query.
- Select Reference.
- Rename the new query as Finance Sales.
- Verify that the new query references the original query.

1.14 Display Column Quality indicators to evaluate data quality.

Solution:

- Open Power Query Editor
- Select the view tab.
- Enable Column Quality
- Review each column's quality indicators.

1.15 Display Column Profile statistics for an imported dataset.

Solution:

- Open Power Query Editor
- Select the View tab
- Enable Column Profile
- Select different columns.
- Review statistical information such as:
 - Count
 - Distinct Values
 - Unique Values
 - Empty Values
 - Error Values
 - Minimum
 - Maximum

1.16 Display the Column Distribution visualization to analyze the frequency of values in a dataset.

Solution:

- Open Power Query Editor
- In the Queries pane, select: Customers_Raw
- Click the View tab
- Enable Column Distribution
- Select the city column
- Observe the frequency distribution.

1.17 Rename multiple queries using a consistent naming convention

Solution:

- Go to the Queries pane
- Right-click Sales_Raw
- Select Rename
- Enter FactSales.
- Repeat for Customers_Raw to DimCustomer
- Repeat for Product_Raw to DimProduct

1.18 Create folders to organize queries inside Power Query Editor

Solution:

- Open Power Query Editor
- Right-click inside the query pane
- Select New Group
- Create groups such as:
 - Staging
 - Dimensions
 - Facts
 - Lookup
 - Archive
- Drag related queries into the appropriate groups.

1.19 Identify transformation errors using the Applied Steps pane.

Solution:

- In the queries, select: Error_Data
- On the right side, find Query Settings to Applied Steps
- Click each Applied step one at a time, starting from the first step
- Look at the Data Preview after each step
- Your goal is to identify the step where an error appears.

1.20 Review the entire Power Query project and verify that it is ready for Data Cleaning and Transformation.

Solution:

- Verify all imported queries
- Review query names
- Confirm query groups
- Verify Data Preview
- Review Applied Steps
- Check Query Dependencies
- Enable data profiling tools
- Verify Formula Bar Visibility
- Confirm Query Settings
- Save your Power BI project.